



Genetic Approaches to Enhance Vermicomposting Efficiency of Earthworms



Giridhar, M¹, Reddy Kumar, A. V.² and Dileep, K. C.¹

¹ Ph. D Scholar, Department of Agronomy, University of Agricultural Sciences, Mandya, Karnataka, India

² Ph. D Scholar, Department of Plant Pathology, University of Agricultural Sciences, Mandya, Karnataka, India

*Corresponding Author's email: girimdhar@gmail.com

Vermicomposting is an eco-biotechnological process in which earthworms and their associated microbiota convert organic wastes into nutrient-rich compost. Process efficiency depends on earthworm physiology, gut-microbial interactions and environmental stress tolerance. Recent advances in genomics, transcriptomics and microbiome research offer new opportunities to enhance key vermicomposting species such as *Eisenia fetida*, *Eudrilus eugeniae* and *Dendrobaena veneta*. This review integrates biological and molecular insights and proposes selective breeding, marker-assisted and genomic selection, and microbiome augmentation as near-term strategies, with functional genomics and gene editing as long-term prospects. Biosafety and socioeconomic implications are also discussed.

Keywords: Vermicomposting, *Eisenia fetida*, Earthworm genetics, Gut microbiome, Selective breeding, Genomic selection, Composting efficiency, Sustainable waste management

Introduction

Organic waste accumulation from agriculture, households and agro-industries poses major environmental and public health challenges. Vermicomposting-bioconversion of these wastes into stable, nutrient-enriched compost through earthworms and associated microbes-offers a sustainable and low-cost solution (Edwards et al. 2021). Earthworms act as natural bioreactors, fragmenting organic residues and stimulating microbial degradation, resulting in humified castings rich in available nitrogen (N), phosphorus (P), potassium (K) and beneficial microbes. Globally, species such as *Eisenia fetida*, *Eudrilus eugeniae* and *Perionyx excavatus* are widely employed due to their high tolerance, fast growth and prolific reproduction. However, composting efficiency is highly variable, influenced by worm genotype, gut microbial composition, substrate quality and environmental factors such as temperature and pH.

Given the increasing demand for rapid, large-scale organic waste processing, researchers are exploring how genetic and molecular enhancement of earthworms can improve performance. The genomic era now allows identification of genes responsible for digestion, detoxification, reproduction and stress tolerance. Combining these insights with selective breeding and microbiome management holds promise for developing “next-generation composting worms.”

The Biological Basis of Vermicomposting Efficiency

Earthworm performance is governed by a set of interacting biological traits:

Feeding and Digestive Efficiency

Feeding rate and enzymatic digestion are primary determinants of compost turnover. Earthworms secrete endogenous enzymes (proteases, cellulases and amylases) while harbouring gut microbes that further catalyse decomposition. Genetic differences influence gut length, enzyme secretion and assimilation efficiency.

Gut Microbial Symbiosis

The earthworm gut hosts dense microbial communities-mainly *Bacillus*, *Pseudomonas* and *Enterobacter* spp.-that mineralize nutrients and suppress pathogens. *E. fetida* gut microbiota have been shown to enrich phosphate-solubilizing and cellulolytic bacteria (Andleeb *et al.* 2022; Hussain *et al.*, 2016). The selective enrichment of these functional groups accelerates humification and nutrient mineralization.

Reproductive Potential

Higher reproductive rates increase biomass and processing capacity. The reproduction rate in *E. fetida* varies with substrate, being highest on cow dung and lowest on lignocellulosic wastes. These differences have genetic underpinnings that can be enhanced through selection.

Stress Tolerance and Detoxification

Earthworms frequently encounter substrates contaminated with metals, pesticides or antibiotics. Genes encoding metallothioneins, heat shock proteins (HSP70) and glutathione S-transferases confer detoxification and tolerance (Homa *et al.* 2005). Heritable resistance observed across generations in *E. fetida* indicates a potential for breeding stress-resistant lines.

Genetic and Species Diversity

Species and strain differences are well documented. For instance, *E. fetida* and *E. andrei* show similar morphologies but distinct composting efficiencies and nutrient output (Jaskulak *et al.* 2022). Such intra- and interspecific diversity provides a foundation for selection and improvement.

Genomic and Molecular Resources

Genome and Transcriptome Datasets

High-throughput sequencing has unveiled genomic blueprints of key species. The *Eudrilus eugeniae* draft genome (488 Mb, ~24,599 genes) provides insights into digestion, stress and reproduction pathways (Arumugaperumal *et al.* 2022). Transcriptomic analyses in *E. fetida* have identified upregulated enzyme and detoxification genes during composting (Huang *et al.*, 2021).

Thousands of microsatellite and SNP markers have been characterized for *E. fetida* and *E. andrei* (Jaskulak *et al.*, 2022), enabling population structure studies, strain identification and marker-assisted breeding.

Functional Gene Expression

Comparative transcriptomic studies under metal stress have shown upregulation of metallothionein and HSP genes, reflecting genomic adaptability. Functional markers for cellulases, ligninases and reproduction hormones are also being identified (Arumugaperumal *et al.* 2022).

Gut Microbiome and Holobiont Function

Earthworm-microbe synergy underpins vermicomposting performance. The gut acts as a selective environment that enriches for beneficial microbes while reducing pathogens. Recent metagenomic analyses (Velásquez-Chávez *et al.*, 2025; Karapantzou *et al.* 2023) revealed that:

- Gut microbiota differs significantly from bulk substrate communities.
- Enriched taxa possess genes for cellulose degradation, nitrogen fixation and phosphorus solubilization.
- Gut passage increases microbial enzymatic potential and nutrient availability in casts.

Field studies integrating earthworms with phosphate-solubilizing bacteria (PSB) enhanced carbon sequestration and nutrient recovery from manure (Wang *et al.* 2022). These results underscore the potential of **holobiont-based selection**-selecting worms together with their beneficial microbial partners.

Materials and Methods / Conceptual Framework

Since this review integrates research synthesis and proposed methodologies, the following approach is recommended for breeding and evaluation of genetically superior earthworm lines:

Population Sampling and Characterization

- Collect distinct earthworm populations (*E. fetida*, *E. eugeniae*, *P. excavatus*) from different ecological niches.
- Confirm species identity via COI DNA barcoding.
- Assess baseline traits: growth rate, cocoon production, substrate consumption, compost nutrient profile.

Selective Breeding Protocol

- Establish breeding cohorts (minimum 50 pairs/line).
- Use truncation selection for high compost conversion efficiency (CCE) and reproductive rate.
- Maintain parallel control lines to monitor genetic drift.
- Continue selection for ≥ 4 generations.

Marker-Assisted Selection (MAS)

- Develop genotyping panels (SSR/SNP markers linked to performance traits).
- Conduct QTL mapping for key traits (e.g., FCE, cocoon rate).

Apply marker-based selection to accelerate gain per generation

Gut Microbiome Profiling

Sequence gut microbiota via 16S rRNA or metagenomics to identify functional microbes. microbial taxa with compost quality and worm growth.

Develop microbial inoculants from beneficial isolates (*Bacillus*, *Pseudomonas*, *Enterobacter*).

Stress Tolerance Assays

- Expose selected lines to sub-lethal heavy metal (e.g., Cu, Zn) or temperature stresses.
- Record survival, feeding rate, metallothionein / HSP70 expression (RT-qPCR).
- Select tolerant genotypes for further breeding.

Evaluation Trials

Conduct replicated trials under diverse substrates (agricultural residues, municipal wastes, paper mill sludge).

Measure:

- Feed Conversion Efficiency (FCE; g waste converted/g biomass/day)
- Compost maturity indices (C: N ratio, humification index)
- Nutrient content (N, P, K)
- Enzyme activities in compost (dehydrogenase, cellulase)

Results and Discussion

Advanced studies have demonstrated significant intra-specific variation in growth and reproduction among *E. fetida* strains. This suggests strong potential for genetic improvement via selection.

Genomic Insights

The *E. eugeniae* draft genome (Arumugaperumal et al. 2022) and *E. fetida* transcriptomes have identified genes for digestive enzymes and stress adaptation, supporting molecular breeding. Genome mining indicates gene families for glycoside hydrolases (GH5, GH9), cellulases and peroxidases involved in lignin breakdown—key targets for enhancing decomposition rate.

Microbiome Contributions

Invermicomposting of cow dung and agro-wastes, *E. fetida* gut microbes increased available P and N by 25-40% compared to non-worm controls (Andleeb et al. 2022). Incorporating PSB inoculants with worms accelerated compost maturity by 20% (Wang et al., 2022). Such findings validate the holobiont enhancement strategy.

Stress Tolerance and Epigenetic Adaptation

E. fetida populations exposed to Cu and Cd stress showed increased metallothionein and HSP expression in F₂ progeny, indicating heritable adaptation or epigenetic memory-an opportunity for breeding resilient strains for industrial waste composting.

Marker Resources and Breeding Tools

Microsatellite libraries (Jaskulak et al. 2022) now allow MAS and population tracking. Combining phenotypic and genotypic data through genomic selection models (GBLUP) could accelerate breeding cycles.

Economic and Ecological Considerations

Genetic improvement can boost vermicomposting throughput by 25-40%, reducing processing time and enhancing nutrient value. However, biosafety and ecological risk assessments must precede field deployment to prevent adverse effects on native fauna and soil ecosystems.

Future Prospects and Research Gaps

1. High-quality chromosome-level assemblies for *E. fetida* and *E. eugeniae* to enhance gene annotation and QTL mapping.
2. Gene-trait association studies for composting efficiency, reproduction and detoxification.
3. Controlled holobiont experiments to quantify co-selection benefits.
4. Exploration of gene editing (CRISPR/Cas) once transformation protocols are optimized.
5. Long-term ecological safety assessments for improved strains.
6. Economic modelling of cost-benefit ratios for breeding programs.
7. Public engagement and extension training to promote safe adoption of improved lines.

Conclusion

Genetic and molecular approaches mark a new frontier in vermicomposting research. Integrating selective breeding, molecular marker tools and microbiome enhancement can substantially improve earthworm composting efficiency, nutrient enrichment and stress resilience. Future incorporation of genomic selection and gene editing, under strict ethical and biosafety frameworks could yield high-performance “bioengineered” earthworm lines for large-scale organic waste management. Such innovations will contribute directly to the goals of circular bio economy, soil health restoration and sustainable agriculture.

References

Andleeb, S., Shafique, I., Naseer, A., Abbasi, W.A., Ejaz, S., Liaqat, I., Ali, S., Khan, M.F., Ahmed, F. and Ali, N.M., 2022. Molecular characterization of plant growth-promoting vermi-bacteria associated with *Eisenia fetida* gastrointestinal tract. *PLoS One*, 17(6), p. e0269946.

Arumugaperumal, A., Sudalaimani, D.K., Arumugaswami, V. and Sivasubramaniam, S., 2022. Draft Genome Sequence of the Earthworm *Eudrilus eugeniae*. *Current Genomics*, 23(2), pp.118-125.

Edwards, C.A., Arancon, N.Q. and Sherman, R.L., 2010. Vermiculture technology: earthworms, organic wastes and environmental management.

Homa, J., Olchawa, E., Stürzenbaum, S.R., Morgan, A.J. and Plytycz, B., 2005. Early-phase immunodetection of metallothionein and heat shock proteins in extruded earthworm coelomocytes after dermal exposure to metal ions. *Environmental Pollution*, 135(2), pp.275-280.

Microbiome Contributions

Invermicomposting of cow dung and agro-wastes, *E. fetida* gut microbes increased available P and N by 25-40% compared to non-worm controls (Andleeb et al. 2022). Incorporating PSB inoculants with worms accelerated compost maturity by 20% (Wang et al., 2022). Such findings validate the holobiont enhancement strategy.

Stress Tolerance and Epigenetic Adaptation

E. fetida populations exposed to Cu and Cd stress showed increased metallothionein and HSP expression in F₂ progeny, indicating heritable adaptation or epigenetic memory-an opportunity for breeding resilient strains for industrial waste composting.

Marker Resources and Breeding Tools

Microsatellite libraries (Jaskulak et al. 2022) now allow MAS and population tracking. Combining phenotypic and genotypic data through genomic selection models (GBLUP) could accelerate breeding cycles.

Economic and Ecological Considerations

Genetic improvement can boost vermicomposting throughput by 25-40%, reducing processing time and enhancing nutrient value. However, biosafety and ecological risk assessments must precede field deployment to prevent adverse effects on native fauna and soil ecosystems.

Future Prospects and Research Gaps

1. High-quality chromosome-level assemblies for *E. fetida* and *E. eugeniae* to enhance gene annotation and QTL mapping.
2. Gene-trait association studies for composting efficiency, reproduction and detoxification.
3. Controlled holobiont experiments to quantify co-selection benefits.
4. Exploration of gene editing (CRISPR/Cas) once transformation protocols are optimized.
5. Long-term ecological safety assessments for improved strains.
6. Economic modelling of cost-benefit ratios for breeding programs.
7. Public engagement and extension training to promote safe adoption of improved lines.

Conclusion

Genetic and molecular approaches mark a new frontier in vermicomposting research. Integrating selective breeding, molecular marker tools and microbiome enhancement can substantially improve earthworm composting efficiency, nutrient enrichment and stress resilience. Future incorporation of genomic selection and gene editing, under strict ethical and biosafety frameworks could yield high-performance “bioengineered” earthworm lines for large-scale organic waste management. Such innovations will contribute directly to the goals of circular bio economy, soil health restoration and sustainable agriculture.

References

Andleeb, S., Shafique, I., Naseer, A., Abbasi, W.A., Ejaz, S., Liaqat, I., Ali, S., Khan, M.F., Ahmed, F. and Ali, N.M., 2022. Molecular characterization of plant growth-promoting vermi-bacteria associated with *Eisenia fetida* gastrointestinal tract. PLoS One, 17(6), p. e0269946.

Arumugaperumal, A., Sudalaimani, D.K., Arumugaswami, V. and Sivasubramaniam, S., 2022. Draft Genome Sequence of the Earthworm *Eudrilus eugeniae*. Current Genomics, 23(2), pp.118-125.

Edwards, C.A., Arancon, N.Q. and Sherman, R.L., 2010. Vermiculture technology: earthworms, organic wastes and environmental management.

Homa, J., Olchawa, E., Stürzenbaum, S.R., Morgan, A.J. and Plytycz, B., 2005. Early-phase immunodetection of metallothionein and heat shock proteins in extruded earthworm coelomocytes after dermal exposure to metal ions. Environmental Pollution, 135(2), pp.275-280.

Hussain, N., Singh, A., Saha, S., Kumar, M.V.S., Bhattacharyya, P. and Bhattacharya, S.S., 2016. Excellent N-fixing and P-solubilizing traits in earthworm gut-isolated bacteria: a vermicompost based assessment with vegetable market waste and rice straw feed mixtures. *Bioresource Technology*, 222, pp.165-174.

Jaskulak, M., Vandenbulcke, F., Rorat, A., Pauwels, M., Zorena, K., Grzmil, P. and Płytycz, B., 2022. Data on the identification of microsatellite markers in *Eisenia fetida* and *Eisenia andrei*. *Data in Brief*, 45, p.108612.

Karapantzou, I., Mitropoulou, G., Prapa, I., Papanikolaou, D., Charovas, V. and Kourkoutas, Y., 2023. Physicochemical changes and microbiome associations during vermicomposting of winery waste. *Sustainability*, 15(9), p.7484.

Velásquez-Chávez, T.E., Sáenz-Mata, J., Quezada-Rivera, J.J., Palacio-Rodríguez, R., Muro-Pérez, G., Servín-Prieto, A.J., Hernández-López, M., Preciado-Rangel, P., Salazar-Ramírez, M.T., Ontiveros-Chacón, J.C. and Peña, C.G.D.L., 2025. Bacterial and Physicochemical Dynamics During the Vermicomposting of Bovine Manure: A Comparative Analysis of the *Eisenia fetida* Gut and Compost Matrix. *Microbiology Research*, 16(8), p.177.

Wang, Z., An, Y., Chen, H., Zhang, J., Zhang, H., Zhu, G., Chen, J., Li, W., Wang, J., Xu, H.J. and Li, Y., 2022. Effects of earthworms and phosphate-solubilizing bacteria on carbon sequestration in soils amended with manure and slurry: a 4-year field study. *Agronomy*, 12(9), p.2064.

